



Finding Market Regimes in Equity Returns Using Hidden Markov Models



Tools & Resources

Software:

- 1** Python
 - 1.1** Yfinance (stock data)
 - 1.2** Numpy/Pandas (data manipulation)
 - 1.3** Matplotlib (plotting)
 - 1.4** HmmLearn (HMM)
 - 1.5** SciPy (visualizations)
- 2** Github (Version Control)

Hardware:

Personal Laptop

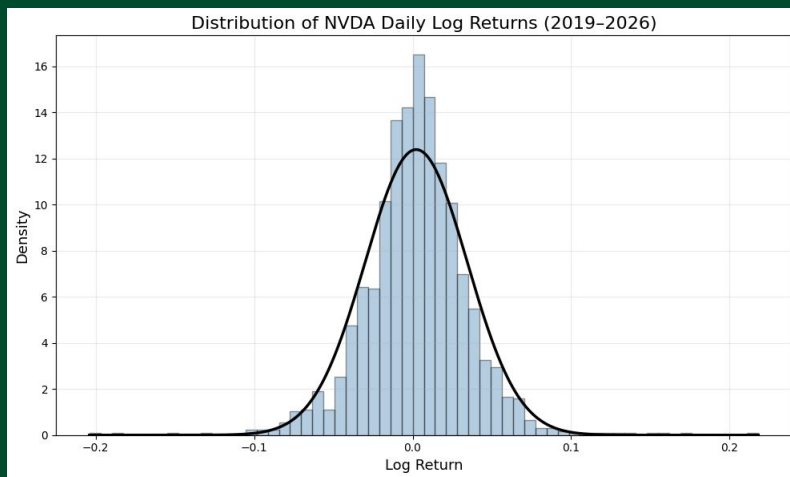
**Advisor/
Consultant:** Professor M. Wade &
Professor M. Gee

Research Question

// Can we automatically detect when the market is calm vs volatile using only price data? //

Nvidia Stock Return

Closing Price



$N(0, \sigma = 0.03)$

From January 1st, 2019 - Feb 18th, 2026

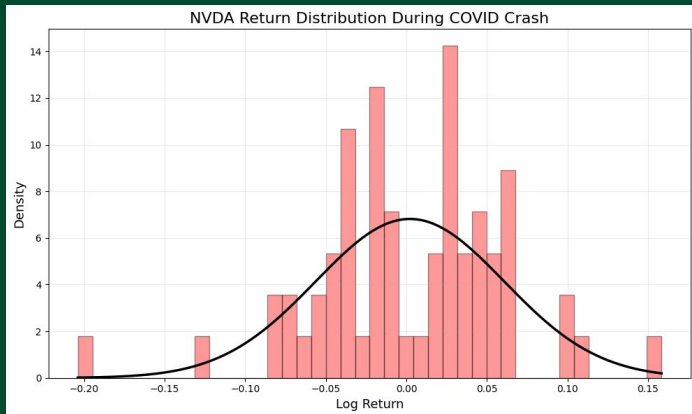
Density = how frequently different return values occur

A few returns are **3+ standard deviations** away → extreme volatility

A **normal distribution** captures the **general** shape

What about when volatility changes?

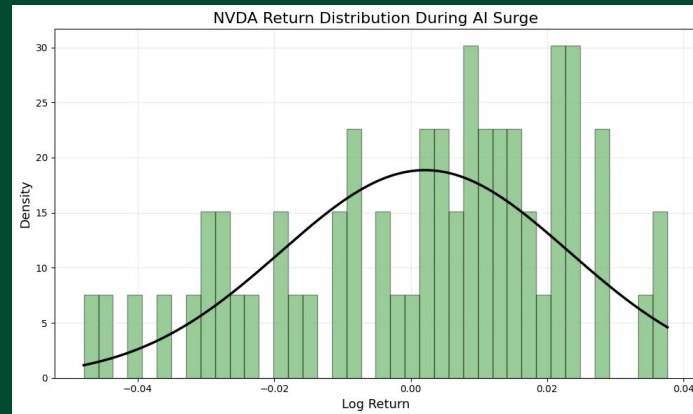
Our Motivation: Markets Change



Covid

Regime

A period where the return follows a **consistent statistical** behavior



AI Surge

Takeaway

This suggests the **market switches** between **volatility regimes** over time



Can a 3-state Hidden Markov Model fitted to daily equity returns identify underlying volatility regimes?

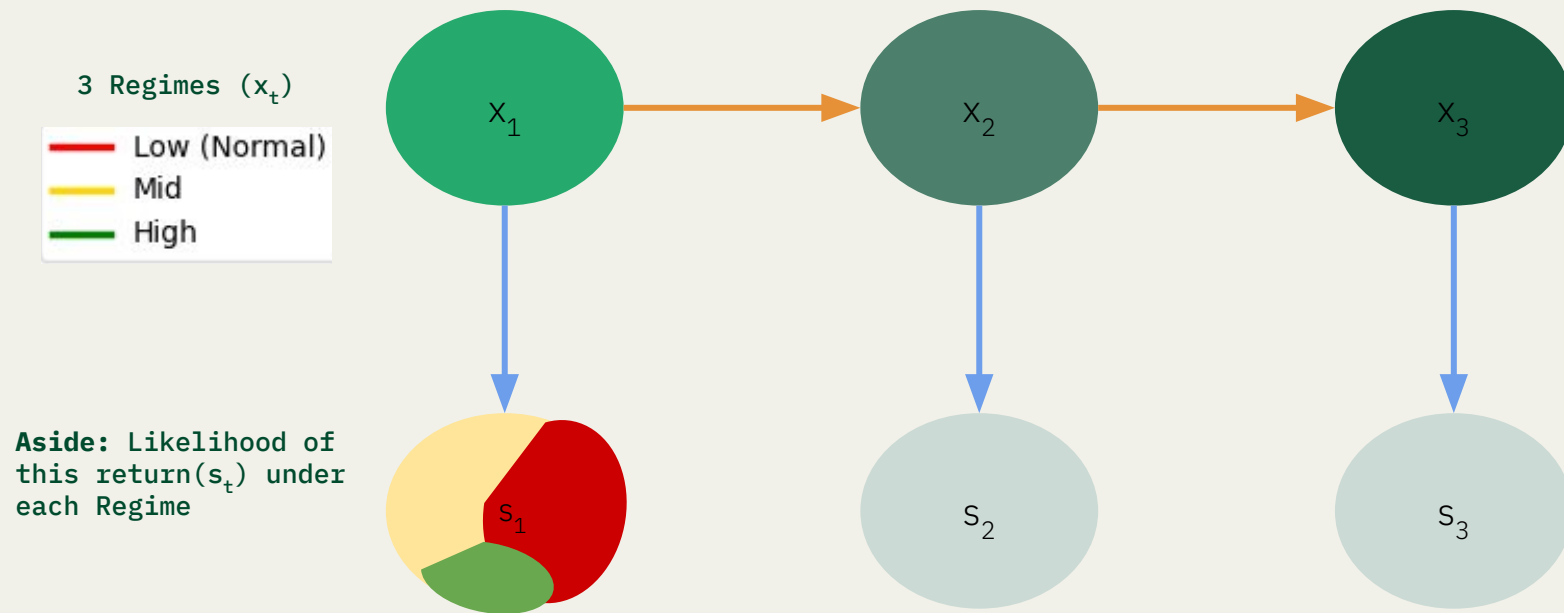
Do those regimes have meaningfully different risk characteristics and "realistic" transition dynamics?



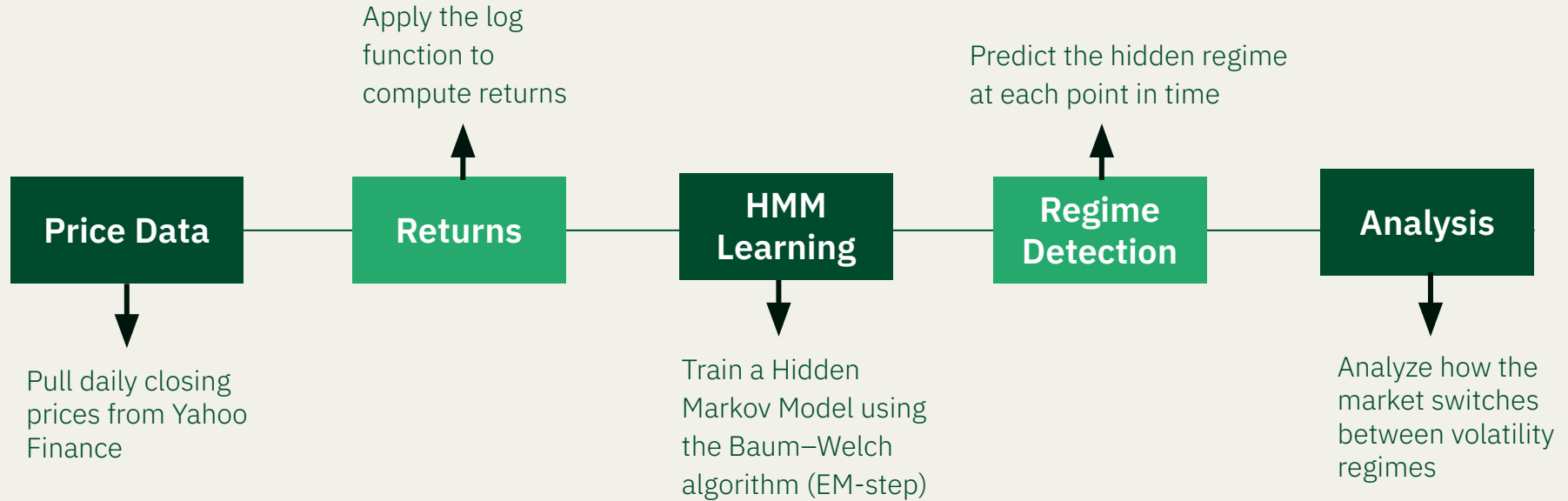
How HMM Solve This

Transition Probability: Probability the Regime change from x_t to x_{t+1}

Emission Probability: Distribution of Returns s_t generated by regime x_t



High Level Process



What the Model Learned

Regime	Volatility	Time Spent (%)
Low	0.022	63%
Med	0.039	36%
High	0.143	1%

Table insights:

- **High-volatility regimes** are **rare** but significantly more extreme
- The **model** identified **three clearly distinct** volatility levels

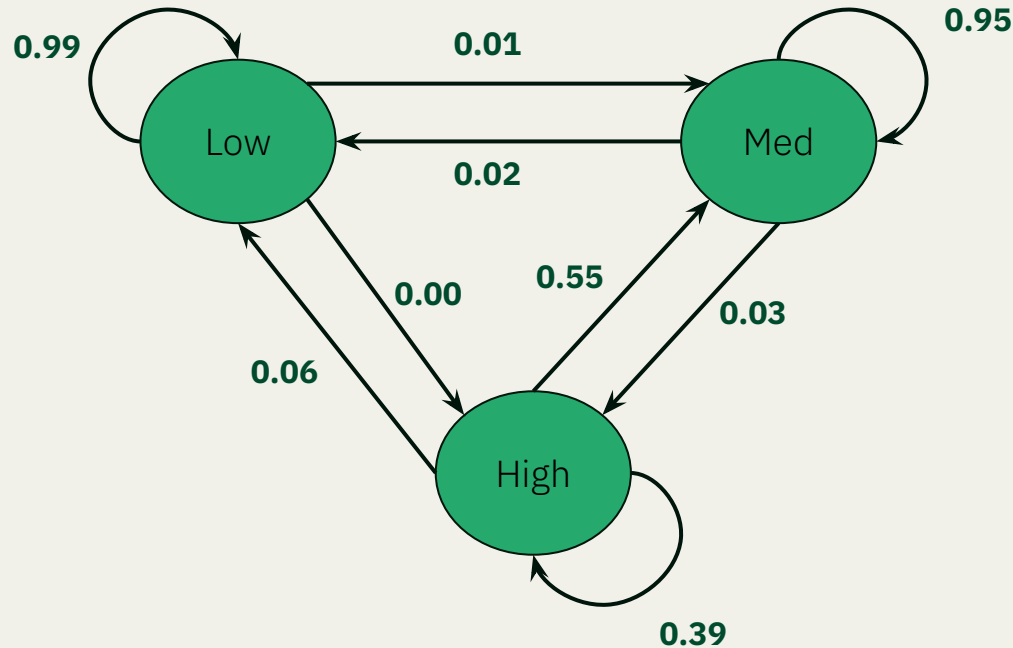
Transition Probability (TP)

	Low	Med	High
Low	0.99	0.01	0.00
Med	0.02	0.95	0.03
High	0.06	0.55	0.39

Matrix Interpretation:

- Rows (i) → **current regime**
- Columns (j) → **next regime**
- Each **entry** shows the **probability of transitioning** from regime i to regime j

Matrix relationship to Graphs



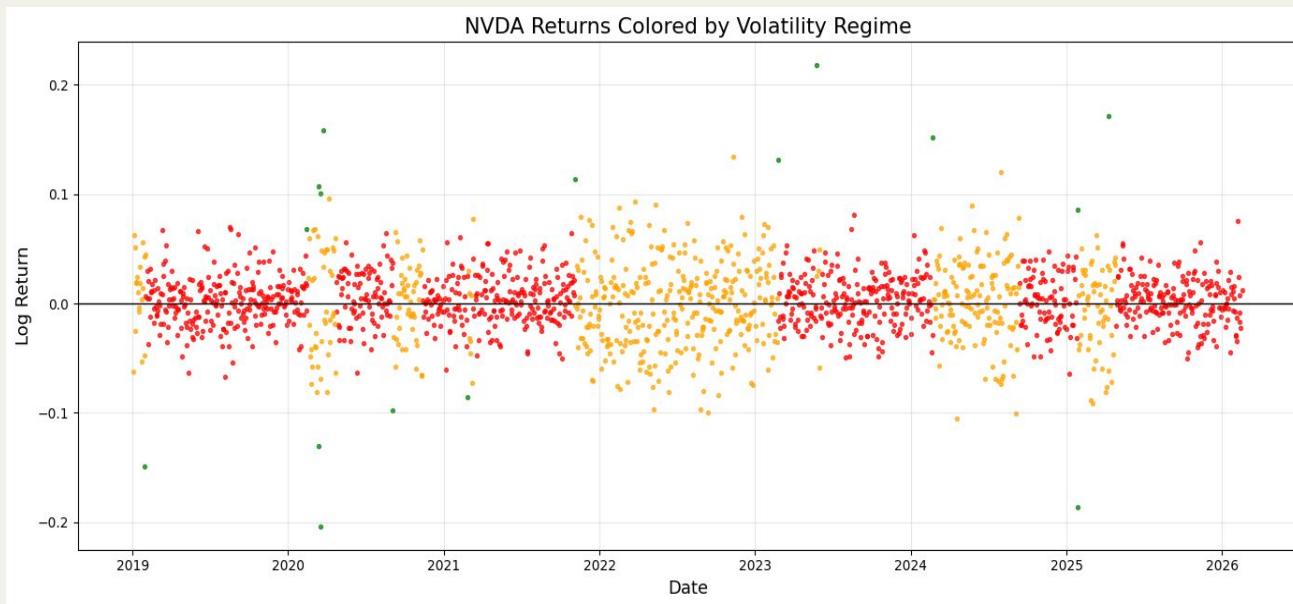
Transition Probability (TP)

	Low	Med	High
Low	0.99	0.01	0.00
Med	0.02	0.95	0.03
High	0.06	0.55	0.39

Matrix Interpretation:

- Rows (i) → **current regime**
- Columns (j) → **next regime**
- Each **entry** shows the **probability of transitioning** from regime i to regime j

Transition Between States



Scatter Plot Observations:

- High-volatility regime occur at **transition periods**
- **Clusters** with med & low volatility

Challenges & Successes

Challenges:

- 01 Global vs Local Optima
- 02 Choosing the number of regimes
- 03 Mapping model output to finance

Successes:

- 01 Trained a HMM
- 02 Detected regimes
- 03 Model converged reliably
- 04 Results were robustly tested



Updated Timeline





Questions?